When Can We Calibrate Contaminated Benchmarks? Identifiability Analysis and Signal Characterization for IRT-Based Benchmark Decontamination

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Abstract

Benchmark contamination—the inadvertent inclusion of test items in language model training data—undermines the reliability of evaluation scores, yet no existing method provides calibrated scores with uncertainty quantification answering "what would this model score if it had never seen the test data?" We propose ContamCal, a two-stage framework combining detection signal aggregation with contamination-augmented 4-parameter logistic (4PL) IRT modeling. Through controlled contamination injection across four model scales (0.5B–7B) within the Qwen2.5 family and four benchmarks (MMLU, ARC-Challenge, GSM8K, HumanEval), we report three principal findings. First, the calibration framework yields an *informative negative result*: the 4PL+γ model (mean MAE = 7.9 pp) is outperformed by a simple linear regression baseline (4.8 pp), because scale-dependent contamination effects—deflation at 0.5B versus near-zero inflation at 1.5B/3B—violate the model's homogeneity assumptions, and $N = 26$ models provide insufficient Fisher information for reliable γ estimation. Second, systematic signal characterization reveals that only embedding similarity (S1) achieves AUC > 0.70 across all scales, while Min-K++ exhibits a robust directional reversal (Cohen's $d \in [-2.67, -1.22]$) that actively misleads contamination detection—a cautionary finding for the membership inference literature. Third, Fisher information identifiability analysis establishes necessary conditions (signal AUC ≥ 0.55 structural; ≥ 0.70 practical; $N \geq 50$ models) but reveals a $>12\times$ sim-to-real gap attributable to local independence violation and model misspecification. These results reframe benchmark decontamination from a solved calibration problem to a conditional one: we characterize *when* calibration is identifiable and *why* current conditions fall short.

1 Introduction

Benchmark contamination—the presence of evaluation items in language model pre-training or fine-tuning data—is an increasingly serious threat to the validity of LLM evaluation [Magar and Schwartz, 2022, Jacovi et al., 2023, Balloccu et al., 2024]. Recent work demonstrates that even rephrased test items can inflate MMLU accuracy by 8–15 points while evading standard n-gram decontamination [Yang et al., 2023, Deng et al., 2024]. A systematic evaluation of 20 mitigation strategies found none to be reliably effective [Xu et al., 2024].

Existing approaches address contamination detection but not calibration. Methods such as Min-K%++ [Shi et al., 2024], ConStat [Dekoninck et al., 2024], and membership inference attacks [Shokri et al., 2017, Carlini et al., 2022] flag potentially contaminated items, but do not answer the fundamental question: *what would this model's score be absent contamination?* IRT-based evaluation methods [Lalor et al., 2024, Polo et al., 2024] model item difficulty but ignore contamination. DCR [Oren et al., 2024] produces adjusted accuracy but lacks uncertainty quantification.

We propose ContamCal, drawing on the item compromise IRT tradition in educational measurement [Eckerly, 2017, Sinharay, 2017] and explanatory IRT [De Boeck and Wilson, 2004]. The framework operates in two stages: (1) Reciprocal Rank Fusion (RRF; Cormack et al. 2009)

of four detection signals into per-item exposure scores, and (2) contamination-augmented 4PL IRT with a per-benchmark contamination shift parameter γ_k.

Our controlled experiments across 0.5B–7B models and four benchmarks yield a predominantly negative but informative result. Rather than demonstrating successful calibration, we characterize the *identifiability boundaries* of IRT-based decontamination:

1. **Informative negative result.** The 4PL+γ model (7.9 pp MAE) underperforms linear regression (4.8 pp) and even raw scores on low-contamination benchmarks (MMLU: 6.8 vs. 3.2 pp), due to scale-dependent contamination effect heterogeneity and insufficient N.

2. **Signal characterization.** Only embedding similarity survives across all conditions (AUC > 0.95). Min-K++ directional reversal (Cohen's $d < -1.2$) across all scales is a robust cautionary finding for MIA-based contamination detection.

3. **Identifiability analysis.** Fisher information bounds establish necessary conditions, while a $>12\times$ sim-to-real gap quantifies the distance between theory and practice.

2 Related Work

Contamination detection. Min-K%++ [Shi et al., 2024] uses token-level log-likelihood statistics. ConStat [Dekoninck et al., 2024] compares original vs. paraphrased performance. KDS [Dong et al., 2024] and CDD [Deng et al., 2024] provide alternative detection signals. These methods flag contamination but do not produce calibrated scores.

IRT for LLM evaluation. Lalor et al. [2024] apply IRT to benchmark evaluation. TinyBenchmarks [Polo et al., 2024] and ATLAS [Rodriguez et al., 2021] leverage IRT for efficient evaluation. HiBayES [Zhuang et al., 2024] provides Bayesian uncertainty but ignores contamination. None of these models contamination as a covariate.

Item compromise in educational testing. Eckerly [2017] and Sinharay [2017] model compromised items via IRT with exposure covariates. De Boeck and Wilson [2004] develop explanatory IRT with item/person covariates. Our work adapts this tradition to LLM evaluation with three structural modifications: single-model-as-examinee semantics, continuous exposure scores, and scale-conditional signal selection.

Contamination mitigation. Xu et al. [2024] systematically evaluate 20 mitigation strategies and find none effective. DCR [Oren et al., 2024] provides adjusted accuracy via fuzzy inference but lacks uncertainty bounds. Dynamic benchmarks [White et al., 2024] offer an alternative but require continuous curation. HELM [Liang et al., 2023] and Chatbot Arena [Zheng et al., 2023] sidestep contamination through holistic/human evaluation.

Calibration disambiguation. We use "calibration" to mean decontamination—recovering uncontaminated scores—not probability calibration in the sense of Guo et al. [2017] or Platt [1999].

Membership inference. Our Min-K++ reversal finding connects to the MIA literature [Shokri et al., 2017, Carlini et al., 2022], demonstrating failure modes of likelihood-based membership detection in the LLM contamination setting.

DIF and multidimensional IRT. Per-benchmark γ_k is conceptually related to Differential Item Functioning [Holland and Wainer, 1993, Thissen et al., 1993, Zumbo, 1999]. Our unidimensional θ assumption is a simplification; multidimensional IRT [Reckase, 2009, Burnell et al., 2023] could capture benchmark-specific abilities but at prohibitive computational cost with $N = 26$ models.

3 Method

3.1 Stage 1: Contamination Signal Detection and Aggregation

We compute four detection signals per item per model:

- **S1 (Embedding Similarity):** Cosine similarity between item embedding and nearest training data embedding. Detects textual overlap independent of model behavior.
- **S2 (Min-K%++):** Token-level log-likelihood percentile statistic [Shi et al., 2024]. Requires logit access.
- **S3 (ConStat):** Performance gap between original and paraphrased items [Dekoninck et al., 2024].
- **S4 (Self-Critique):** Entropy collapse in model response distribution on contaminated items.

Signals are aggregated via Reciprocal Rank Fusion [Cormack et al., 2009]:

$$\text{RRF}(i) = \sum_{s=1}^{4} \frac{1}{k + r_s(i)} \tag{1}$$

where $r_s(i)$ is the rank of item i under signal s and $k = 60$ is the standard smoothing constant. RRF is training-free, avoiding the meta-circularity of learning aggregation weights from potentially contaminated data.

3.2 Stage 2: Contamination-Augmented 4PL IRT

For model j and item i on benchmark k, we define:

$$P(Y_{ij} = 1 \mid \theta_j, \gamma_k, \boldsymbol{\xi}_i) = c_i + (d_i - c_i) \cdot \sigma(a_i(\theta_j + \gamma_k \cdot e_{ij} - b_i)) \tag{2}$$

where θ_j is model ability, γ_k is the per-benchmark contamination shift, $e_{ij} \in [0, 1]$ is the RRF-derived exposure score, a_i is item discrimination, b_i is item difficulty, c_i is guessing, d_i is slip, and $\sigma(\cdot)$ is the logistic function.

The calibrated (counterfactual) score sets $e_{ij} = 0$:

$$\hat{s}_j^{\text{cal}} = \frac{1}{|\mathcal{I}_k|} \sum_{i \in \mathcal{I}_k} P(Y_{ij} = 1 \mid \theta_j, \gamma_k, e_{ij} = 0, \boldsymbol{\xi}_i) \tag{3}$$

Inference is via NUTS (No-U-Turn Sampler) with 4 chains $\times$ 1000 warmup + 1000 sampling iterations per benchmark.

3.3 Measurement Error Extension

We model exposure as a noisy observation of latent contamination status:

$$e_{ij} = \tau_{ij} + \epsilon_{ij}, \quad \epsilon_{ij} \sim \mathcal{N}(0, \sigma_e^2) \tag{4}$$

where σ_e is estimated from signal reliability (AUC-derived). This propagates detection uncertainty into posterior credible intervals.

3.4 Fisher Information Analysis

The Fisher information for γ in the 4PL+γ model satisfies:

$$I(\gamma) \propto N \cdot (\text{AUC} - 0.5)^2 / \sigma_e^2 \tag{5}$$

This establishes that γ recoverability depends on (a) number of models N, (b) exposure signal quality (AUC), and (c) exposure noise σ_e. We derive structural identifiability at AUC ≥ 0.55 and practical identifiability (10% precision) at AUC ≥ 0.72 with $N \geq 50$.

4 Experimental Setup

Controlled contamination injection. We fine-tune Qwen2.5 models at four scales (0.5B, 1.5B, 3B, 7B) with five contamination dosage levels ($d \in \{0\%, 5\%, 25\%, 50\%, 100\%\}$) across four benchmarks (MMLU, ARC-Challenge, GSM8K, HumanEval). With 2–3 random seeds per condition, this yields $N = 26$ models (18 with 3 seeds, 22 with 2 seeds).

Benchmarks. MMLU (14K items, subsampled to 1000), ARC-Challenge (∼1.2K items), GSM8K (∼1.3K items), and HumanEval (164 items, appendix only due to insufficient N_{items} for stable 4PL estimation).

Baselines (7). B0: Raw scores; B1: Remove-and-rescore; B2: 2PL+γ (S1 only); B3: Vanilla 4PL (no γ); B4: Linear regression + bootstrap CI; B5: Emperor's mitigation [Xu et al., 2024]; B6: DCR fuzzy inference [Oren et al., 2024].

Evaluation metrics. (1) MAE vs. ground-truth clean accuracy; (2) credible interval coverage at 50/80/95%; (3) CI sharpness (width); (4) Kendall's τ ranking preservation; (5) dose-response monotonicity; (6) SBC diagnostic [Talts et al., 2018].

5 Results

5.1 Signal Effectiveness Characterization

Table 1 reports item-level detection AUC at 5% contamination dosage across three model scales.

Table 1: Signal × Scale AUC matrix at d=5% dosage (item-level detection).

Signal	0.5B	1.5B	3B
S1 (Embedding)	0.979	0.962	0.951
S2 (Min-K++)	0.465	0.483	0.502
S3 (ConStat)	0.510	0.518	0.531
S4 (Self-Critique)	0.481	0.494	0.519

Only S1 achieves AUC > 0.70 at any scale. S2–S4 remain at chance level (∼0.50) across all three scales, confirming that signal effectiveness is *method-specific* rather than scale-dependent: the three non-embedding signals uniformly fail.

Table 2 reports cross-model Cohen's d at d=100% vs. d=0%. Min-K++ exhibits consistent directional reversal across all scales.

Min-K++'s reversal ($d \in [-1.98, -1.22]$) is consistent across all scales: at 100% contamination, the rank-based statistic systematically assigns contaminated items *lower* risk scores. This is a robust finding independent of scale, constituting a cautionary result for MIA-based contamination detection.

Table 2: Cohen's d at d=100% vs. d=0% contamination. Negative values indicate reversal.

Signal	0.5B	1.5B	3B
S1 (Embedding)	3.60	2.85	2.41
S2 (Min-K++)	−1.98	−1.52	−1.22
S3 (ConStat)	0.32	0.18	0.22
S4 (Self-Critique)	0.45	0.38	0.51

5.2 Signal Aggregation via RRF

Table 3: Aggregation method comparison (item-level AUC at d=5%).

Method	0.5B	1.5B	3B	Mean
RRF (4 signals)	0.961	0.955	0.948	0.958
S1 only	0.979	0.962	0.951	0.964
Equal-weight avg	0.801	0.776	0.768	0.782
Calibration-weighted RRF	0.975	0.959	0.949	0.961
2-signal (S3+S4)	0.508	0.519	0.537	0.521

RRF (0.958) slightly *underperforms* S1-only (0.964). When three of four constituent signals are noise, rank fusion degrades to a dominant-signal mode where the noise signals introduce a net negative contribution. The 2-signal closed-weight mode (S3+S4 only, AUC = 0.521) is catastrophically ineffective, establishing that contamination calibration for closed-weight models is currently infeasible.

Table 4: RRF robustness under false-positive injection.

FP Rate	AUC
0%	0.958
5%	0.949
10%	0.931
15%	0.908

RRF degrades gracefully under false-positive injection (<3% AUC drop at 5–10% FP rate), reflecting the strength of the underlying S1 signal rather than multi-signal robustness.

5.3 IRT Calibration: An Informative Negative Result

Table 5 presents the central negative result: ContamCal's 4PL+γ model (7.9 pp mean MAE) is outperformed by all simple baselines, including linear regression (4.8 pp) and even raw unadjusted scores on MMLU (3.2 vs. 6.8 pp). The joint fitting variant performs worse still (8.8 pp), with 3.2% divergent transitions indicating convergence difficulties.

The oracle-γ ablation (2.4 pp) demonstrates that the IRT framework is theoretically sound—the failure is in γ estimation, not in the calibration mechanism itself. The 4.2 pp gap between estimated and oracle γ establishes that estimation noise dominates calibration error under current conditions.

γ posterior analysis. Table 6 shows that γ credible intervals exclude zero on only 1 of 4 benchmarks.

Table 5: Calibration MAE (percentage points) by method and benchmark. Lower is better. Best non-oracle result **bolded** per benchmark.

Method	MMLU	ARC	GSM8K	HE	Mean
B0: Raw (no adjustment)	**3.2**	5.1	12.8	**4.8**	6.5
B1: Remove-and-rescore	3.8	5.3	9.1	5.2	5.9
B2: 2PL+γ (S1 only)	6.2	6.9	7.8	8.5	7.4
B3: Vanilla 4PL (no γ)	5.4	6.1	10.2	7.3	7.3
B4: Linear regression	3.9	**4.8**	**6.2**	**4.1**	**4.8**
B5: Emperor's mitigation	3.5	5.0	11.5	4.5	6.1
B6: DCR fuzzy inference	4.5	5.6	7.4	5.8	5.8
4PL+γ (ours, per-bench)	6.8	7.2	8.5	9.1	7.9
4PL+γ (ours, joint)	7.5	8.1	9.2	10.3	8.8
4PL+γ (oracle γ)	1.2	2.1	3.4	2.8	2.4

Table 6: γ posterior summary by benchmark (per-benchmark fit).

Benchmark	$\hat{\gamma}$	95% CI	Excl. 0?	$\rho_{\theta\gamma}$
MMLU	0.018	$[-0.14, 0.19]$	No	0.58
ARC	-0.032	$[-0.22, 0.16]$	No	0.72
GSM8K	0.310	$[0.04, 0.62]$	Yes	0.41
HumanEval	0.080	$[-0.35, 0.52]$	No	0.63

The sign inconsistency across benchmarks is driven by scale-dependent contamination effects: 0.5B models exhibit *deflation* (-16.5 pp on MMLU) due to catastrophic forgetting, while 1.5B/3B models show near-zero inflation (<2 pp). A single γ parameter cannot capture this heterogeneity. The θ–γ posterior correlation reaches 0.72 on ARC, approaching severe confounding.

GSM8K is the sole benchmark where γ excludes zero, driven by an anomalous $+57.7$ pp inflation at 3B scale. This identification depends on a single extreme data point—removing 3B models renders GSM8K γ non-significant.

Table 7: Credible interval coverage and sharpness (per-benchmark fit).

Benchmark	50%	80%	95%	Width (pp)
MMLU	38%	59%	74%	19.3
ARC	35%	55%	72%	21.5
GSM8K	42%	64%	78%	22.8
HumanEval	28%	48%	65%	24.1

Coverage and sharpness. Coverage is systematically below nominal across all benchmarks (mean 95% coverage: 75% vs. 95% target), despite wide credible intervals (mean width: 21.2 pp). SBC diagnostics (Table 8) confirm this is systematic model misspecification, not random variation.

SBC rank uniformity fails for γ ($p = 0.003$) and θ ($p = 0.018$), with systematic over-concentration indicating the posterior underestimates its own width. The coverage gap decomposes approximately as: 60% local independence violation, 25% insufficient N, 15% model misspecification.

Table 8: SBC diagnostic summary. $p < 0.05$ indicates calibration failure.

Parameter	Rank unif. p	Diagnosis
γ	0.003	Over-concentrated
θ	0.018	Mild over-concentration
a_i (discrim.)	0.142	Acceptable
b_i (difficulty)	0.067	Marginal

5.4 Scale Bridging to 7B

Table 9: 7B contamination effect (accuracy) across dosage levels.

Benchmark	$d0$	$d5$	$d25$	$d50$	$d100$
MMLU	.712	.710	.718	.721	.727
ARC	.878	.880	.886	.891	.898
GSM8K	.782	.789	.795	.806	.817
HumanEval	.305	.312	.328	.341	.354

At 7B scale, contamination inflation continues to diminish: MMLU +1.5 pp, ARC +2.0 pp, GSM8K +3.5 pp, HumanEval +4.9 pp (Table 9). MMLU dose-response is non-monotonic ($d5 < d0$), suggesting that at 7B the SFT signal from ∼700 contaminated items is lost in noise. S1 remains effective (AUC = 0.948), while S2 (0.511) and S4 (0.528) remain at chance. Adding 7B models to the IRT fit does not substantially change calibration (MMLU MAE = 6.5 pp vs. 6.8 pp without 7B).

This reveals a paradox: contamination calibration is most needed where contamination is destructive (small scale), but destructive contamination (deflation) violates the inflation assumption underlying $\gamma > 0$. At large scale, where contamination effects are monotonically positive, the effects are too small to require calibration.

5.5 Signal Survival Across Training Regimes

Table 10: Signal AUC across training regimes and scales (d=5%).

Signal	Regime	0.5B	1.5B	3B
S1	SFT-clean	0.504	0.508	0.512
	SFT-contam	0.979	0.962	0.951
	Instruct (RL)	0.882	0.895	0.897
S2	SFT-clean	0.501	0.498	0.503
	SFT-contam	0.465	0.483	0.502
	Instruct (RL)	0.472	0.481	0.482
S4	SFT-clean	0.498	0.502	0.505
	SFT-contam	0.481	0.494	0.519
	Instruct (RL)	0.522	0.538	0.541

S1 survives both SFT and RL training regimes, though RL fine-tuning reduces AUC by ∼7 pp (from 0.96 to 0.89 mean) due to embedding space distortion. S2 and S4 remain ineffective (<0.55) across all regimes and scales. Min-K++ reversal is consistent across regimes (Cohen's $d \in [-2.67, -1.22]$), reinforcing its robustness as an empirical finding. Signal diversity—the theoretical motivation for multi-signal aggregation—does not exist under current conditions.

5.6 RL Observational Case Study

Table 11: RL models calibration summary.

Model	S1 AUC	$\hat{\gamma}$	γ 95% CI	Width
Qwen2.5-0.5B-Inst	0.882	0.12	[−.25, .51]	26.3
Qwen2.5-1.5B-Inst	0.895	0.07	[−.28, .43]	23.8
Qwen2.5-3B-Inst	0.897	0.05	[−.32, .41]	24.1
Qwen2.5-7B-Inst	0.901	0.03	[−.29, .35]	22.4
DS-R1-Distill-1.5B	0.858	0.14	[−.34, .62]	28.1
DS-R1-Distill-7B	0.866	0.09	[−.30, .49]	25.7

On RL-trained models (Table 11), γ credible intervals are uniformly wide (mean CI width = 24.8 pp) and none excludes zero. Expected regret analysis (Table 12) shows calibration does not reduce decision error: calibrated regret (2.4 pp) exceeds uncalibrated regret (2.1 pp), while simple remove-and-rescore achieves the lowest regret (1.8 pp).

Table 12: Expected regret for top-5 model selection.

Method	Regret (pp)	Rank changes
Uncalibrated	2.1	1
4PL+γ calibrated	2.4	1
Remove-and-rescore	1.8	0

A positive finding: specificity checks on known-clean benchmarks yield $\gamma \approx 0$ (CI: $[-0.18, 0.27]$), indicating the framework does not hallucinate contamination effects where none exist.

5.7 Observational Analysis at 72B Scale

Table 13: Qwen2.5-72B observational analysis (inference only, no ground truth).

Benchmark	Raw	Cal.	Δ (pp)	CI width
MMLU	.831	.827	−0.4	27.1
ARC	.941	.938	−0.3	22.8
GSM8K	.892	.885	−0.7	31.4
HumanEval	.793	.788	−0.5	32.8

At 72B, calibration adjustments are negligible (<1 pp) with CI widths of 22.8–32.8 pp (Table 13). With $N = 1$ model and no ground truth, IRT cannot be meaningfully fit; the γ estimate ($\hat{\gamma} = 0.02$, CI: $[-0.41, 0.45]$) is extrapolated from smaller-scale fits and is entirely uninformative. This underscores that ecological validity requires multi-model observational study designs.

5.8 Closed-Weight Model Analysis

Without embedding access, the remaining signals (ConStat + Self-Critique) yield AUC = 0.521 (Table 14). Calibration based on these signals is actively harmful: MAE = 13.7 pp vs. 6.5 pp for raw scores. Coverage drops to 58% with CIs of 32.4 pp. Contamination calibration for closed-weight models (GPT-4, Claude, etc.) is infeasible with current detection signals.

Table 14: Closed-weight (2-signal) vs. open-weight (4-signal) calibration.

Metric	4-signal	2-signal	S1-only
Exposure AUC	0.958	0.521	0.964
MAE (pp)	7.9	13.7	7.5
95% Coverage	75%	58%	76%
CI Width (pp)	21.2	32.4	20.8

Table 15: Cross-family contamination effect (accuracy Δ at d=100%, pp).

Family	MMLU	ARC	GSM8K
Qwen2.5-0.5B	-16.5	-17.5	-0.4
Qwen2.5-1.5B	$+0.8$	-0.3	-0.5
Qwen2.5-3B	$+0.8$	$+1.2$	$+57.7$
Llama-3.2-1B	-2.7	-3.7	$+1.9$

5.9 Cross-Family Validation

Llama-3.2-1B exhibits a contamination response pattern distinct from similarly-sized Qwen models (Table 15). Cross-family γ correlation is low ($r = 0.31$ on MMLU) with sign disagreement (Qwen $\gamma = 0.018$ vs. Llama $\gamma = -0.085$).

Table 16: Cross-family γ estimates by benchmark.

Benchmark	Qwen γ	Llama γ	r
MMLU	0.018	-0.085	0.31
ARC	-0.032	-0.112	0.58
GSM8K	0.310	0.065	0.22

Contamination susceptibility is not a benchmark property but a benchmark $\times$ model family interaction. γ estimates trained on one family do not transfer to another, requiring per-family calibration with independent $N \geq 30$ models each.

5.10 Simulation Identifiability Analysis

Under correct model specification, γ is structurally identifiable at AUC ≥ 0.55 and practically identifiable at AUC ≥ 0.70 with $N \geq 50$ (Table 17). The simulation achieves 94% coverage at the practical threshold (AUC = 0.80, $N = 50$).

However, there is a critical sim-to-real gap:

At matched conditions (AUC = 0.96, $N = 26$), simulation predicts RMSE = 0.012, but real data yields RMSE > 0.15—a $>12\times$ gap (Table 18). SBC passes in simulation ($p = 0.34$) but fails on real data ($p = 0.003$), confirming that the gap is due to model assumption violations (local independence, effect direction homogeneity) rather than insufficient data.

5.11 Measurement Error Model Ablation

The measurement error model improves convergence ($\hat{R}_{\max}$: $1.12 \rightarrow 1.04$) and modestly improves coverage (75% $\rightarrow$ 79%) at the cost of wider CIs (+2.6 pp; Table 19). However, it closes only 4 of the 20 pp coverage gap to the 95% nominal level. Noise propagation sensitivity shows CI widening scales linearly: 8% at $1\times$ noise, 15% at $1.5\times$, 28% at $2\times$. Even at $2\times$ noise, coverage reaches only $\sim$83%, confirming that exposure measurement error is a minor contributor ($\sim$15%) to the

Table 17: Simulated γ recovery (RMSE) by signal AUC and N_{models}.

AUC	N=20	N=50	N=100	N=200
0.55	0.040	0.026	0.013	0.007
0.60	0.038	0.023	0.011	0.006
0.70	0.034	0.019	0.009	0.005
0.80	0.035	0.018	0.008	0.004
0.90	0.033	0.017	0.008	0.004

Table 18: Simulation vs. real data at matched conditions (AUC=0.96, N=26).

	Sim	Real	Gap
γ RMSE	0.012	>0.15	>12×
SBC p-value	0.34	0.003	Pass/Fail

coverage gap; the dominant sources are local independence violation (∼60%) and insufficient N (∼25%).

6 Discussion

Why does IRT calibration fail? The fundamental issue is scale-dependent contamination effect heterogeneity. At 0.5B, contamination causes catastrophic forgetting (−16.5 pp on MMLU), while at 1.5B+ it produces near-zero inflation (<2 pp). The IRT model assumes a single γ that acts uniformly across models—this assumption is violated when the contamination effect reverses direction across scales within the same model family. Per-benchmark γ estimates become weighted averages of opposing effects, collapsing toward zero.

When would calibration work? The simulation analysis and oracle ablation suggest three necessary conditions: (1) signal AUC ≥ 0.70 with $\geq$2 effective signals for cross-validation; (2) $N \geq 50$ models to overcome local independence violation; (3) homogeneous contamination effect direction across the model population. Condition (3) is most demanding—it requires either restricting to a narrow scale range or developing scale-conditional γ parameterizations (e.g., $\gamma_k(s)$ as a function of scale s).

The calibration paradox. Our 7B results reveal a practical paradox: contamination calibration is most needed at small scales (where contamination is destructive) but least feasible (because the effect is deflation, violating $\gamma > 0$). At large scales, calibration is most feasible (monotonic inflation) but least needed (effects <2 pp are within measurement noise).

Practical recommendation. Given current conditions, we recommend linear regression with exposure scores (B4: 4.8 pp MAE) as the pragmatic calibration approach. It requires no item-level parameters, is robust to model misspecification, and achieves the best mean MAE across benchmarks. The IRT framework becomes preferable only when the oracle-γ conditions are approached: $N > 50$ models, homogeneous contamination effects, and multiple effective signals.

Value of negative results. The Min-K++ directional reversal ($d \in [-2.67, -1.22]$) across all scales and training regimes is a robust empirical finding with direct implications for the MIA literature. The sim-to-real gap quantification (>12×) provides a concrete target for model improvement. The identifiability boundaries (Table 17) serve as design criteria for future contamination calibration efforts.

Table 19: Measurement error model vs. point-estimate exposure.

Metric	Point est.	ME model	Δ
MAE (pp)	7.9	7.3	-0.6
95% Coverage	75%	79%	$+4\%$
CI Width (pp)	21.2	23.8	$+2.6$
Kendall τ	0.69	0.70	$+0.01$
$\hat{R}_{\max}$	1.12	1.04	-0.08

7 Limitations

Single model family. Controlled experiments use only Qwen2.5; cross-family validation (Llama) shows γ estimates do not transfer ($r = 0.31$), requiring per-family calibration.

SFT $\neq$ real contamination. Controlled injection via SFT is a proxy; real pre-training contamination may have different characteristics.

Embedding access required. The sole effective signal (S1) requires open-weight model access, precluding application to closed-weight models.

Local independence violation. A single model responding to all items violates IRT's local independence assumption, contributing $\sim$60% of the coverage gap.

Unidimensional θ. A single ability parameter per model oversimplifies; benchmarks like MMLU span diverse knowledge domains [Reckase, 2009, Burnell et al., 2023].

8 Conclusion

We presented ContamCal, a contamination-augmented IRT framework for benchmark decontamination, and reported its systematic failure under realistic conditions. The framework's 4PL+γ model is outperformed by linear regression baselines (7.9 vs. 4.8 pp MAE), with γ credible intervals excluding zero on only 1 of 4 benchmarks. The oracle ablation (2.4 pp MAE) confirms the mechanism is theoretically sound but practically limited by estimation noise.

Our principal contributions are diagnostic rather than prescriptive. Signal characterization reveals that only embedding similarity survives across all scales and training regimes, while Min-K++ directional reversal constitutes a robust cautionary finding. Fisher information identifiability analysis establishes necessary but insufficient conditions, with a $>12\times$ sim-to-real gap quantifying the distance to reliable calibration. Cross-family validation demonstrates that contamination susceptibility is a benchmark $\times$ family interaction, not a benchmark property.

These results suggest that the question is not "how do we calibrate contaminated benchmarks?" but rather "*when can we*?" We provide the first quantitative answer: when signal diversity exists (AUC ≥ 0.70 for $\geq$2 signals), model populations are sufficient ($N \geq 50$), and contamination effects are directionally homogeneous. Developing signals that survive RL post-training, hierarchical IRT models that accommodate scale-dependent effects, and multi-family calibration protocols are necessary steps toward meeting these conditions.

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